

Search for Events in XENON1T associated with Gravitational Waves

Various
Various

(XENON Collaboration)

We perform a blind search for events in the XENON1T dark matter detector that occur close in time to gravitational waves observed in LIGO. No events are observed in the nuclear recoil, electronic recoil, CE ν NS, and S2-only channels within ± 500 seconds of gravitational wave events. This constrains theories Beyond the Standard Model that predict relativistic particles being emitted during binary mergers.

I. INTRODUCTION

Following the first detection of a gravitational wave (GW) signal by the Laser Interferometer Gravitational-Wave Observatory (LIGO) [1], signals from binary mergers are now routinely detected by the LIGO/Virgo observatories [2]. The subsequent observation of electromagnetic counterparts from binary neutron star coalescence [3] has made GW events an integral part of multi-messenger astrophysics. Various particle detectors have conducted follow-up searches for events associated with GW events in a time window of $\pm [400\text{--}500]$ s of the observed GW signal, which might indicate the simultaneous emission of relativistic particles such as neutrinos. The sensitive (neutrino) energy ranges of these searches, summarized in Table I, span from EeV down to half a MeV. No evidence of a coincident signal has been reported yet. We use data from the XENON1T dark matter search [4] to extend this energy window down to 0.19 keV, which is particularly interesting to constrain signals from physics Beyond the Standard Model, such as those expected from relativistic axions [5] or sterile neutrinos [6, 7].

TABLE I. Previous searches for events coincident with GW and the energy ranges they are sensitive to.

Detector	Detection energy
Pierre Auger Observatory [8]	100 PeV – 25 EeV
IceCube [9]	100 GeV – 100 PeV
Super-Kamiokande [10]	3.5 MeV – 100 PeV
XMASS [11]	14 MeV – 100 MeV
KamLAND [12]	1.8 MeV – 111 MeV
NOvA [13]	1 MeV – 10 MeV
Borexino [14]	0.5 MeV – 5 MeV

II. DATA SETS

The LIGO/Virgo observatories conducted their second observation run from November 30, 2016, to August 25, 2017 [20]. Overlapping this time frame, XENON1T took Science Run 0 (SR0) data from November 22, 2016, to January 18, 2017, and Science Run 1 (SR1) data from February 2, 2017, to February 8, 2018. This data has already been analyzed in several channels that we revisit

here to check for events coincident with LIGO/Virgo observations. A search for nuclear recoils (NR) demanding both scintillation light (S1) and electroluminescence from ionization electrons (S2) is available during both SR0 [15] and SR1 [16]. In addition, using data from SR1, a search was conducted for electronic recoils (ER) [17], for coherent elastic neutrino-nucleus scattering (CE ν NS) [18], and for nuclear recoils where the requirement of a scintillation signal has been waived (S2-only) [19].

Details of these analysis channels can be found in the respective papers [15–19]. Notably, the NR and ER channels demand coincident signals from at least three photomultiplier tubes (PMTs) for an S1, while the CE ν NS channel requires signals from only two PMTs. Furthermore, the S2 threshold has been reduced from 200 photoelectrons (PE) and 500 PE for the NR and ER channels, respectively, to 120 PE and 150 PE for the CE ν NS and S2-only channels, respectively. The corresponding energy ranges of these channels are given in Table II in units of keV electron-equivalent (keV_{ee}) for the ER and S2-only channels, and as keV nuclear recoil-equivalent (keV_{nr}) for the NR and CE ν NS channels. Effective xenon target masses vary for different analyses. Whereas the fiducial volume limits the NR, ER, and CE ν NS channels, the S2-only channel is limited by the strong event selections applied to reduce backgrounds.

Background rates in the NR, CE ν NS, and S2-only channels are only a few events (14, 6, and 170, respectively) observed during the entire Science Runs. This motivates the use of a single energy bin for each of these channels. From Figure 3 of [17], we can see that the ER spectrum is not constant in energy. The spectrum exhibits spectral lines due to ^{83m}Kr and ^{131m}Xe . As can be seen in Table III, we define five energy bins for the ER channel: low energy (0–30 keV), ^{83m}Kr (30–50 keV), medium energy (50–142 keV), ^{131m}Xe (142–185 keV) and high energy (185–210 keV).

Three of the eight GW events that LIGO/VIRGO observed during SR0 and SR1 occurred during XENON1T calibration or downtime. Only SR0 NR data is available during GW170104, while SR1 data in all four analysis channels is available during GW170729, GW170817, GW170818, and GW170823.

TABLE II. Characteristics of considered analysis channels used in this study. For each of these channels, we show the corresponding energy ranges, the Science Run data that have been analyzed for that channel, considered fiducial target mass, and the observed event rate. Further, in the ± 500 s time window around a GW event, we give the average livetimes for detecting a signal, the average number of expected events (N_{exp}), and the probability of observing zero events ($P(0)$).

Event channel	Energy range	Runs analyzed	Fiducial target mass	Event rate (events/day)	Livetime (per GW)	N_{exp}	$P(0)$
NR [15, 16]	4.9 keV _{nr} – 40.9 keV _{nr}	SR0, SR1	1.3 tonne	0.05	935 s	0.00054	0.9995
ER [17]	1 keV _{ee} – 210 keV _{ee}	SR1	1.042 tonne	186	935 s	See Table III	See Table III
CE ν NS [18]	1.6 keV _{nr} – 40.9 keV _{nr}	SR1	1.04 tonne	0.03	790 s	0.00027	0.9997
S2-only [19]	0.19 keV _{ee} – 3.7 keV _{ee}	SR1	0.121 tonne	0.94	650 s	0.0071	0.993

III. BACKGROUND ESTIMATION

Consistent with previous coincidence searches shown in Table I, that used a time window of $\pm [400-500]$ s, we use a ± 500 s time window centered on the GW event time. This is appropriate for any relativistic particle flux emitted by the binary merger. Accounting for deadtime due to various cuts and background events, livetimes for detecting a signal in the ± 500 s time window around a GW event in the NR, ER, CE ν NS and S2-only channels are approximately 935 s, 935 s, 790 s and 650 s respectively. Background rates are assumed to be constant for the duration of the Science Runs for the NR, CE ν NS, and S2-only channels. Therefore, for these channels,

$$\text{event rate} = \frac{\text{total events during entire Science Run}}{\text{total livetime of entire Science Run}}. \quad (1)$$

However, as can be seen in Figure xxx of [reference the Figure from time-stability paper], this is a poor approximation for the ER channel. We thus use the same background estimation for the ER channel as in [cite time-stability paper]. Before unblinding this data, we compared the values predicted by these background models with the observed values for all ER channels using the $(-1500, -500)$ s sideband before the actual GW events. Given five independent ER energy bins and four GW events, we divide the chi-square by 20 to find the reduced chi-square statistic to be 1.14, confirming that these background models agree with the observation. In the ± 500 s time window around a GW event, the expected number of background events is calculated as

$$N_{\text{exp}} = \text{event rate} * \frac{\text{livetime}}{24 * 60 * 60}. \quad (2)$$

The corresponding probabilities of observing zero events are values of cumulative distribution functions at zero, assuming Poisson statistics. These values are given in Table II for the NR, CE ν NS, and S2-only channels and in Table III for the ER channel.

While the ± 500 s time window was still blinded during background estimation, we calculated an upper limit on the number of expected events in the ER channel, based on the background model, above which evidence for a signal could have been claimed. We used Poisson

statistics with a trial factor t_f of 20 and applied the Sidak Correction [21]:

$$p_{\text{global}} = 1 - (1 - p_{\text{local}})^{1/t_f}. \quad (3)$$

The local p-value of 0.003 corresponds to a global p-value of 0.00015 for a potential excess at a 3σ confidence level. The resulting upper limits on the expected background events are between 2 and 5 events, as shown in Table III. We have not given upper limits on the number of expected events in other channels because values of cumulative distribution functions at 1, for Poisson distributions with such small means, are more than 99.99%.

IV. RESULTS AND DISCUSSION

Once these detection channels, time and energy ranges, and background models were fixed, event timing information was unblinded. No event was observed in any of these channels within ± 500 seconds of any of the tested GW events. Given the low background rates, this is expected under the background-only hypothesis in the NR, CE ν NS, and S2-only channels. The total number of expected background events in the ER channel across all GW events is ~ 3.4 . However, even in this channel, no events were observed, consistent with the background-only hypothesis, assuming Poisson statistics, within a 3σ confidence interval.

We have thus provided the first search for signals in the keV energy region associated with GW events. Despite the 1.3 tonne fiducial mass of XENON1T, energy regions going down to 0.19 keV, and multiple nuclear or electronic recoil channels, no signals were found. The detection or a lack of sub-keV particles within ± 500 s time windows around GW events allows for a better understanding of the particles-interaction in binary mergers. This generally holds for any particle with a coupling to xenon, be it Standard Model neutrinos or particles from theories Beyond the Standard Model, such as the sterile neutrinos and Axions.

V. ACKNOWLEDGEMENTS

We are grateful...

TABLE III. Predictions for the different energy bins of the ER channel during four GW events. For each bin, in the ± 500 s time window around a GW event, we give the number of expected events (N_{exp}), the probability of observing zero events ($P(0)$) and the upper limits (UL) on the expected background events at a 3σ confidence level.

	Low Energy			^{83m}Kr			Medium Energy			^{131m}Xe			High Energy		
	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL	N_{exp}	$P(0)$	UL
GW170729	0.058	0.94	2	0.16	0.85	3	0.22	0.80	3	0.64	0.52	5	0.07	0.93	2
GW170817	0.058	0.94	2	0.15	0.86	3	0.22	0.80	3	0.29	0.75	4	0.07	0.93	2
GW170818	0.058	0.94	2	0.15	0.86	3	0.22	0.80	3	0.27	0.76	4	0.07	0.93	2
GW170823	0.058	0.94	2	0.14	0.87	3	0.22	0.80	3	0.23	0.79	3	0.07	0.93	2

- [1] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. Lett. **116**, 061102 (2016), 1602.03837.
- [2] R. Abbott et al. (LIGO Scientific, VIRGO, KAGRA) (2021), 2111.03606.
- [3] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. Lett. **119**, 161101 (2017), 1710.05832.
- [4] E. Aprile et al. (XENON), Eur. Phys. J. C **77**, 881 (2017), 1708.07051.
- [5] D. F. G. Fiorillo and F. Iocco, Phys. Rev. D **105**, 123007 (2022), 2109.10364.
- [6] M. U. Frensel, Ph.D. thesis, Basel U. (main) (2017).
- [7] M. Frensel, M.-R. Wu, C. Volpe, and A. Perego, Phys. Rev. D **95**, 023011 (2017), 1607.05938.
- [8] E. Santos (Pierre Auger), J. Phys. Conf. Ser. **1181**, 012060 (2019).
- [9] R. Hussain, J. Vandenbroucke, and J. Wood (IceCube), PoS **ICRC2019**, 918 (2020), 1908.07706.
- [10] Y. Nakano (Super-Kamiokande), Universe **5**, 7 (2018).
- [11] K. Abe et al. (XMASS), Astropart. Phys. **129**, 102568 (2021), 2007.16046.
- [12] A. Gando et al. (KamLAND), Astrophys. J. Lett. **829**, L34 (2016), 1606.07155.
- [13] M. A. Acero et al. (NOvA), Phys. Rev. D **101**, 112006 (2020), 2001.07240.
- [14] M. Agostini et al. (BOREXINO), Astrophys. J. **850**, 21 (2017), 1706.10176.
- [15] E. Aprile et al. (XENON), Phys. Rev. Lett. **119**, 181301 (2017), 1705.06655.
- [16] E. Aprile et al. (XENON), Phys. Rev. Lett. **121**, 111302 (2018), 1805.12562.
- [17] E. Aprile et al. (XENON), Phys. Rev. D **102**, 072004 (2020), 2006.09721.
- [18] E. Aprile et al. (XENON), Phys. Rev. Lett. **126**, 091301 (2021), 2012.02846.
- [19] E. Aprile et al. (XENON), Phys. Rev. Lett. **123**, 251801 (2019), 1907.11485.
- [20] B. P. Abbott et al. (LIGO Scientific, Virgo), Phys. Rev. X **9**, 031040 (2019), 1811.12907.
- [21] Z. Šidák, Journal of the American Statistical Association **62**, 626 (1967).